

ARNAU NECHES VILÀ

Seville, Spain

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EDUCATION

The University of Sydney – Sydney, Australia

Aug. 2025 – Jun. 2026

Exchange Student, Mathematics & Computer Science | GPA: 93/100 | **Top 5%**

Selected coursework: Natural Language Processing, Deep Learning, Operating Systems Internals, Differential Geometry, Nonlinear ODEs with Applications, Complex Analysis, Financial Derivatives

Universidad de Sevilla – Seville, Spain

Sep. 2022 – Expected May 2027

Double Degree in Mathematics & Computer Science | GPA: 9.3/10 | 78+ ECTS awarded with **highest honours**

Selected coursework: Real & Complex Analysis, Probability Theory, Abstract Algebra, Mathematical Programming, Algorithms & Data Structures, Artificial Intelligence, Computer and Network Architecture

RESEARCH EXPERIENCE

MINERVA AI Lab, Universidad de Sevilla – Research Assistant

Jul. 2025 – Present

- Developed and evaluated two new feature-selection methods designed to better capture interactions between features; led the research, implementation and experiments across **6 synthetic and 4 real-world datasets**.
- First author** of “Lookahead and Dynamic Beam Search Variants of FOCI for Conditional-Dependence Feature Selection,” accepted at **CAEPIA 2026** and forthcoming in Lecture Notes in Artificial Intelligence (LNAI).
- Co-authored “Benchmarking Chronos-2 as a Foundation Model for Traffic Forecasting” also accepted at CAEPIA 2026 and forthcoming in LNAI.

PROFESSIONAL EXPERIENCE

Welyto – Lead Software Engineer

Apr. 2026 – Aug. 2026

Early-stage startup | *Conversational AI device for older adults*

- Developed an AI-powered conversational device for older adults as part of a two-engineer team, spanning backend architecture, cloud infrastructure, AI integration, real-time communication and the family-facing application.
- Took the product from development to deployment with **50 paying beta families** using physical devices, supporting concurrent conversations, persistent user memories and preferences, and family-to-device messaging.
- Built the backend and device-cloud platform using Python, FastAPI, PostgreSQL, WebSockets, Google Cloud, Vertex AI and React, with the server orchestrating AI interactions, tools, memory and real-time communication.

SELECTED PROJECT

Accessible Activity Data Platform – University of Sydney

Aug. 2025 – Oct. 2025

Project Manager & Developer | **7-person team** | *Client: Institute of Musculoskeletal Health*

- Led a 7-person team over a 13-week capstone for the Institute of Musculoskeletal Health, serving as project manager, developer, and client liaison to build a platform helping people with disabilities discover suitable physical activities.
- Built an AI-powered data collection pipeline that discovered activity providers online and transformed unstructured web content into structured data; developed the database and web interface and integrated Gemini API and Google Maps API.

LEADERSHIP & ACTIVITIES

Competitive Programming Club – Universidad de Sevilla

2024 – Present

Events & Institutional Relations

- Active competitive programmer; qualified with my team for the national **Ada Byron** programming contest in both 2024 and 2025.
- Organize academic and industry events and manage external relations, building relationships with **10+ companies** for sponsorships, talks and collaborations.

Sputnik / Los 300

2025

- Selected for Los 300 after completing Sputnik, a selective programme bringing together young people from southern Spain around technology, entrepreneurship, innovation, leadership and social impact.

SKILLS

Programming: Python, C, Java, JavaScript, SQL, MATLAB

ML & Data: PyTorch, scikit-learn, Pandas, NumPy

Software: FastAPI, React, PostgreSQL, SQLAlchemy, WebSockets, REST APIs

Cloud & Tools: Google Cloud Platform, Docker, Git, Linux

Languages: Spanish (Native), English (C1; Cambridge Advance, Grade A), Catalan (Native)